AI PRODUCT MANAGEMENT · CASE STUDY

CognArc

Agentic Cognitive-Behavioral AI Evaluation & Alignment Platform

Always-On · Human Oversight Built-In · Powered by TRIBE v2

9	54	3	4
Feature Pillars	Total Features	Build Phases	Oversight Zones

Role	AI Product Manager · Co-Founder
Product Type	Agentic AI Evaluation Platform
Architecture	Always-on cognitive agent with structured human oversight
Research Base	TRIBE v2 — tri-modal foundation model; 1,000+ hrs fMRI; 720 subjects
Scope	Zero-to-one product design: problem framing, architecture, feature strategy, roadmap, GTM
Market	AI evaluation tooling · Cognitive safety · Agentic AI governance

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01 · EXECUTIVE SUMMARY

What CognArc Is and Why It Exists

CognArc is an always-on cognitive agent that continuously evaluates AI systems and digital experiences for cognitive impact — surfacing load, confusion, trust erosion, and manipulation across every team that ships AI-powered products.

Five different teams inside a modern AI company share a common problem they have never had a shared tool for. The AI engineer needs to know whether a prompt change made outputs harder for users to understand — before it ships. The PM needs to know whether the onboarding flow is cognitively overloading users — before they churn. The growth team needs to know which landing page variant will actually be understood — before they spend media budget. The designer needs cognitive evidence for their prototype — before user research begins. The red team needs to know whether a model update re-introduced a manipulation pattern they previously fixed — before it reaches users again.

Each of these teams currently works blind or reactive. They discover cognitive problems through complaints, churn, failed campaigns, and post-hoc research. CognArc gives all five teams a shared cognitive intelligence layer — powered by TRIBE v2, a foundation model trained on over 1,000 hours of fMRI data across 720 subjects, capable of predicting how the human brain responds to any text, audio, or visual stimulus — that runs continuously, without instigation, across every surface they own.

> **The design premise:**
> *Cognitive quality is not a property any single team owns. It is a property the entire product either maintains or degrades — in every output, every interface, every campaign, every model update. The only architecture that can maintain it continuously across all of those surfaces is an agent. Not a tool five teams remember to call. An agent that never stops watching.*

The platform is built around a non-negotiable principle: autonomy must be proportional to consequence. CognArc monitors and scores freely. It recommends with full evidence. It acts autonomously only where the stakes are low. For everything consequential — blocking a production output, deploying a rewrite, filing a regulatory report — it prepares a complete decision package and waits for explicit human approval. The Trust Gradient Engine enforces this structurally, not by convention.

As the PM and co-founder on this project, I designed CognArc from the problem outward: identifying five distinct buyer problems that share a single underlying cause, building the oversight architecture before defining a single feature, and then constructing nine feature pillars that span AI output evaluation, behavioral analytics integration, marketing and growth intelligence, UI/UX cognitive testing, manipulation and trust detection, guardrail enforcement, AI eval platform integration, benchmarking, and red team augmentation. This case study documents that thinking end to end.

Five Teams, One Shared Blind Spot

The Measurement Problem Nobody Has Named

The AI industry has a measurement problem, and it is not the one most people discuss. The dominant frameworks for evaluating AI outputs — accuracy, faithfulness, toxicity scores, hallucination rates — measure the output in isolation. They tell you whether the AI got the answer right. They tell you nothing about what that answer does to the person receiving it.

This gap does not only affect AI safety teams. It affects every team that ships anything an AI system touches. The engineer who changed a prompt does not know whether it made outputs harder to understand. The growth team that spent $80K on a campaign does not know whether low conversion is a targeting problem or a comprehension problem. The designer who rebuilt the onboarding flow does not know which step is cognitively overloading users until churn data surfaces three weeks later. The red team that fixed a manipulation pattern last quarter does not know whether the latest model update re-introduced it. None of these teams have a cognitive measurement instrument. All of them need one.

Five Problems, One Underlying Cause

The Team	The Problem They Cannot Currently Solve	What Happens Without CognArc
AI Engineer	No way to know whether a prompt or model change made outputs cognitively worse before it ships	Cognitive regressions discovered by users. Reported as confusion or drop-off in analytics. Root cause never identified.
Product Manager	No cognitive visibility into onboarding, activation, or retention flows until behavioral data accumulates	Churn attributed to friction. The cognitive cause — overload, comprehension failure, trust timing mismatch — remains invisible.
Growth & Marketing	No way to test whether creative, copy, or campaigns are understood before media spend	Low conversion misdiagnosed as targeting failure. Comprehension failure goes unfixed. Budget wasted on the wrong optimization.
Product Designer	No cognitive evidence for design decisions before user research. Prototype problems found too late.	User research spent discovering problems that could have been simulated. Design iteration starts from user feedback, not cognitive evidence.
Red Team / AI Safety	No continuous manipulation monitoring. Red team findings are	Manipulation patterns re-introduced by model updates go undetected. Safety posture asserted but not auditable.

The Team	The Problem They Cannot Currently Solve	What Happens Without CognArc
	periodic, unverified, and hard to defend to regulators.	

Why Scale Makes Every One of These Worse

When AI was deployed slowly and cautiously, reactive discovery was tolerable. A team could review major changes, run periodic evals, and catch problems before they compounded. That window has closed. Modern AI products ship prompt changes daily, model updates weekly, and UI variations continuously. A single LLM-powered application generates millions of user-facing outputs per hour. A marketing team runs dozens of variants simultaneously. A design team ships flow changes across multiple surfaces at once.

At that scale, three structural failures become unavoidable with any reactive approach:

- **Humans evaluate what they think to submit. Every problem they have not specifically looked for is invisible — until it surfaces as damage.** Coverage collapse.
- **By the time a cognitive problem is detected, reported, and fixed, the exposure is already measured in millions of outputs, thousands of users, and weeks of compounding harm.** Latency as damage.
- **As AI surfaces multiply across engineering, growth, design, and safety functions, the evaluation workload grows faster than any team's headcount. Triage is inevitable. The evaluations that do not happen are where the problems hide.** Evaluation burden that cannot scale.

The Full Scope of What Current Tools Miss

Failure Mode	Why Current Tools Miss It	How CognArc Addresses It
Gradual trust erosion across sessions	Only visible in longitudinal user research — weeks after it begins	Trust Coherence monitored per session longitudinally. Significant decline detected and escalated within 7 days.
Manipulation drift in LLM outputs	Caught only if someone schedules a manipulation evaluation	Continuous scanner scores every output against 6-category taxonomy the moment it is generated. No trigger needed.
Subgroup cognitive risk hidden in aggregate	Requires deliberate analyst segmentation — aggregate metrics hide it	Population-Level Variance Engine continuously flags outputs harmful to specific cohorts without instigation.
Onboarding load accumulation	Found in user research weeks post-ship, after dropout has already occurred	Onboarding Load Curve Agent generates cognitive load profile automatically on every flow update, before any user sees it.

Failure Mode	Why Current Tools Miss It	How CognArc Addresses It
Prompt cognitive regression	Only caught if an engineer runs a regression eval after each model update	Autonomous Prompt Regression Monitor detects cognitive decline the moment a new model version is connected.
Comprehension failure in marketing copy	Inferred from post-launch conversion data — indirect signal, arrives too late	Comprehension Confidence scored pre-launch. Growth teams test before spend, not after.
Creative cognitive quality pre-spend	No instrument exists to evaluate cognitive impact of creative before media budget is committed	Creative Cognitive Evaluator scores any ad, email, or landing page before a dollar is spent on media.
Red team finding regression post-fix	No continuous monitoring after remediation. Model updates can silently re-introduce fixed patterns	Continuous manipulation scanner detects re-emergence of previously remediated patterns the moment a model update ships.
UI trust timing mismatch	Discovered in user interviews, weeks after onboarding is live	Trust Timing Analyzer flags steps that ask for trust before it has been earned, before any user encounters them.

Why Agentic Is the Only Logical Architecture

Every one of these failure modes has the same underlying structure: a cognitive problem that emerged gradually, was invisible to reactive tools, and was only discovered after it had already done damage. A tool you call when you suspect a problem cannot prevent any of them. It can only respond to what you already know to look for.

An agent that monitors every output, every flow, every campaign asset, and every model update continuously — across all five buyer contexts simultaneously — is not a premium version of a reactive tool. It is the only architecture that actually matches the nature of the problem.

> **The framing that defined every design decision:**
> *Cognitive quality is not a property you check for. It is a property you maintain. You cannot maintain something you only inspect occasionally. Maintenance requires continuous monitoring. Continuous monitoring, across five buyer contexts and millions of daily outputs, requires an agent.*

The Science That Made This Possible

An agentic cognitive evaluation platform is only credible if its underlying science is sound. The foundational research question I had to answer before designing a single feature was: is there a scientifically validated model capable of predicting human cognitive response to arbitrary stimuli at production scale? The answer, as of 2025, is yes. That model is TRIBE v2.

What TRIBE v2 Is

TRIBE v2 is a tri-modal foundation model trained to predict human brain activity in response to naturalistic stimuli — video, audio, and language. Published in 2025, it was trained on over 1,000 hours of fMRI data across 720 subjects. Its defining capability is generalization: it accurately predicts brain responses for novel stimuli, novel tasks, and novel subjects, surpassing traditional linear encoding models by several-fold.

Critically, TRIBE v2 was validated against seminal experimental paradigms from cognitive neuroscience — visual attention tasks, neuro-linguistic comprehension tasks, multisensory integration tasks — and recovered results that took decades of empirical research to establish. This is not a black-box prediction model. Its outputs are interpretable: specific brain regions, specific cognitive processes, specific activation patterns that map to recognisable human experiences like confusion, trust, and emotional response.

How I Productized TRIBE

Understanding TRIBE as a research system was the first task. Understanding what it would take to run it at production scale as a continuous perceptual layer for an AI agent was the harder one.

From tool to sensory system	In a reactive product, TRIBE would be a tool you call when you need a score. In CognArc, TRIBE is the agent’s sensory system — always running, continuously perceiving the cognitive landscape of every connected output stream. This reframing changes the infrastructure requirements entirely.
Inference at production latency	The full TRIBE model is computationally expensive. Serving continuous evaluation at sub-600ms latency requires model distillation: training a smaller student model on TRIBE’s outputs to approximate its predictions at a fraction of the compute cost. The distilled model handles real-time continuous scoring. The full model runs asynchronously for deep analyses.
Domain shift mitigation	TRIBE was trained on naturalistic stimuli — films, spoken language, natural scenes. Digital UI artifacts and LLM outputs are domain-shifted inputs. Phase 1 is scoped to text and image modalities where this gap is smallest, with a hold-out validation set of digital stimuli used to measure and monitor domain-shift risk before Phase 2 expansion.

Feedback loop architecture	As CognArc accumulates validated behavioral data post-launch, that data can fine-tune TRIBE’s simulation accuracy. This compounding loop — where the agent gets smarter as it operates — is the platform’s long-term moat. It is also the highest-risk operation in the system, and requires the strictest human oversight. The fine-tuning pipeline is Phase 3, with a hard human gate at every stage.
The interpretability requirement	A cognitive AI agent that produces unexplained scores would be scientifically indefensible and commercially unsaleable. Every CognArc output — every score, flag, and recommendation — exposes the top contributing brain regions, the confidence interval, and a plain-language explanation of what the score means. No black-box conclusions. This is non-negotiable.

04 · THE PRODUCT

What CognArc Is

The One-Sentence Definition

> *CognArc is an always-on cognitive agent that gives every team shipping AI-powered products — engineers, PMs, growth, design, and safety — a shared, continuous, neuroscientifically grounded view of whether their work is actually working with human cognition or against it.*

What It Is Not

CognArc is not an analytics platform. It does not replace Amplitude or Mixpanel. It sits one layer above them, giving behavioral data the cognitive interpretation it currently lacks. It is not a content moderation tool — content moderation classifies what is in an output; CognArc classifies what that output does to the person receiving it. It is not a UX research service, a marketing optimization tool, or a red-teaming firm. It is the continuous cognitive intelligence layer that makes all of those functions more effective by giving them access to predictions they currently have no way to generate.

Five Buyers, One Platform

CognArc is designed from the ground up to serve five distinct buyer types within a single product architecture. Each has a different workflow, a different definition of value, and a different entry point. The platform unifies them because the underlying problem is the same: cognitive impact is invisible without the right instrument, and no team currently has one.

Buyer	Their Core Problem	What CognArc Gives Them
AI Engineer	Cognitive regressions in prompts and model updates are invisible to accuracy-only eval	Autonomous Prompt Regression Monitor + CI/CD Cognitive Gate. Continuous detection. No scheduling required.
Product Manager	No cognitive visibility into onboarding, activation, or retention flows until behavioral data accumulates post-launch	Onboarding Load Curve Agent + Cognitive-Behavioral Alignment Score. Pre-launch and continuous post-launch.
Growth & Marketing	No instrument to test whether creative, copy, or campaigns are understood before media spend	Creative Cognitive Evaluator + Variant Ranker + Cognitive Funnel Mapper. Test before spend, not after.

Buyer	Their Core Problem	What CognArc Gives Them
Product Designer	No cognitive evidence for design decisions until user research — which arrives after the prototype is already built	Zero-Traffic A/B Engine + Prototype Heatmap + Comprehension Gap Detector. Evidence before user research, not from it.
Red Team / AI Safety	Manual, periodic testing at a fraction of the output volume. Findings unverified post-fix. Regulatory evidence trail incomplete.	Continuous manipulation scanner + Trust Erosion Monitor + Immutable Audit Log. Coverage at scale, evidence at depth.

The go-to-market motion leads with AI engineers and PMs — they have budget authority, move fast, and generate the usage data that makes the growth, design, and regulatory offerings credible. Growth and design teams are the expansion motion within accounts. Red team and compliance functions are the enterprise and regulatory motion.

Competitive Position

CognArc vs.	Their Position	CognArc's Specific Edge
Braintrust / Langfuse / Arize	Task-performance eval: accuracy, faithfulness, hallucination. Human-triggered runs.	Always-on cognitive impact scoring. Detects what accuracy metrics cannot see. No human trigger needed.
Datadog / Arize LLM monitoring	Monitors output accuracy, latency, and model drift.	Monitors cognitive impact on users — what outputs do to the person, not what they are as data.
Amplitude / Mixpanel	Behavioral analytics: what users did and when.	Cognitive interpretation: why they did it. Write-back enriches existing analytics with cognitive labels.
Content moderation APIs	Classifies what is in the content: toxicity, hate, explicit material.	Classifies what the content does cognitively: manipulation, confusion, trust erosion.
Optimizely / A/B testing tools	Requires statistical traffic to determine a winner. Post-launch discovery.	Cognitive simulation determines the winner before launch. No traffic required.
UX research firms	Qualitative, retrospective, expensive, non-scalable.	Quantitative, prospective, scalable, pre-launch via TRIBE simulation.
AI red-teaming services	Periodic adversarial testing by human experts. No continuous monitoring post-fix.	Continuous manipulation detection at scale. Regression monitoring after every model update.

05 · THE ARCHITECTURE

How the System Works

The architecture of CognArc was shaped by a constraint that most AI platforms do not have: it needs to serve five different buyer workflows simultaneously, continuously, without any of them needing to trigger it. An engineer's CI/CD pipeline, a growth team's creative upload, a designer's Figma workspace, a PM's analytics dashboard, and a red team's safety monitoring function all need to receive cognitive intelligence from the same underlying agent — in real time, in the format each team already works in, without requiring any of them to learn a new tool.

That constraint produced three architectural decisions that define the platform: a two-tier inference model that keeps continuous scoring fast while preserving deep analysis capability, an event-driven pipeline that connects CognArc to every buyer surface without polling, and a Trust Gradient that governs every agent action so that human oversight is structural rather than procedural. Everything else follows from those three decisions.

The Continuous Intelligence Loop

The agent operates as a closed loop with seven stages. Every stage serves a buyer function — not just an infrastructure function. The loop runs continuously without human instigation except where the Trust Gradient requires approval.

1 · Sense	Agent ingests AI outputs, prompt changes, UI deployments, creative uploads, and analytics events from all connected systems in real time. No human trigger. An engineer's merged PR, a designer's Figma export, a marketer's asset upload — all become inputs automatically.
2 · Perceive	Distilled TRIBE model scores every stimulus within 600ms: Cognitive Load, Comprehension Confidence, Emotional Valence, Trust Coherence, Manipulation Risk. This is the perceptual layer — the moment CognArc understands what a stimulus will do to the person receiving it.
3 · Detect	Scores compared against baselines, configured thresholds, and population-level models. A regression in an engineer's prompt. A comprehension drop in a designer's onboarding step. A trust erosion trend in a campaign. All surface as anomalies the moment they cross a threshold.
4 · Reason	Agent generates root cause analysis, counterfactual alternatives, and ranked remediation options. The engineer gets a rewrite recommendation. The designer gets a resequencing suggestion. The growth team gets a ranked variant list. The red team gets a manipulation evidence package.
5 · Act or Escalate	Trust Gradient determines what happens next. Low stakes — a PR comment, a dashboard flag, a Slack alert: agent acts and logs. High stakes — a production

	block, a prompt deployment, a regulatory report: agent surfaces a full decision package and waits for human approval.
6 · Validate	Post-action, the agent monitors whether the intervention worked. TRIBE's prediction is reconciled against real behavioral signals from the SDK and analytics connectors. Divergence is surfaced. The PM sees whether the cognitive fix produced the behavioral improvement.
7 · Learn	Validated behavioral data is structured as TRIBE training examples and queued for fine-tuning. The agent's simulation accuracy compounds over time. Hard human gate before any fine-tuning run executes — this is Phase 3, behind the strictest governance in the system.

How the System Serves Each Buyer Surface

The same underlying architecture delivers different value to each buyer, through different entry points, without any of them needing to understand the full system.

Buyer Surface	Entry Point	How the Architecture Serves Them
AI Engineer	CI/CD plugin + Eval API	Every PR touching prompts or model config triggers autonomous evaluation. Cognitive scores appear alongside accuracy scores in their existing eval platform. Regressions flagged before merge.
Product Manager	Behavioral SDK + Analytics connectors	Every user session is cognitively labeled in real time. Cognitive-behavioral alignment score surfaces where TRIBE predictions diverged from real behavior. Analytics write-back means cognitive data lives in Amplitude, not a new dashboard.
Growth & Marketing	Asset upload + Creative Evaluator	Every uploaded creative triggers automatic TRIBE scoring. Variant rankings returned within 5 minutes. Brand trust drift monitored longitudinally. Cognitive funnel maps generated on a weekly cadence without scheduling.
Product Designer	Figma integration + Prototype upload	Every Figma frame tagged for review generates an automatic heatmap. Every onboarding flow update generates an automatic load curve. Comprehension gaps and trust timing mismatches flagged before user research begins.
Red Team / Safety	Continuous scanner + Audit log	Every output scored against the manipulation taxonomy automatically. Trust erosion monitored per session longitudinally. Immutable audit log provides regulatory-grade evidence trail for every flag and every remediation.

System Components

TRIBE Inference Layer	Two-tier: distilled model for continuous synchronous scoring (<600ms p95); full model on async GPU cluster for counterfactual generation, population variance modeling, benchmark suite, and regulatory audits. Stimulus hash caching prevents redundant inference at scale.
Behavioral SDK + Connectors	Lightweight JS + mobile SDK (<8KB gzipped) capturing cognitive friction signals. Bidirectional sync with Segment, Amplitude, Mixpanel, PostHog, GA4. Real-time cognitive labeling of existing event streams. Write-back of cognitive scores as custom properties into existing analytics platforms.
Guardrail Engine	CI/CD plugins (GitHub Actions, GitLab CI, Jenkins) + Prompt Evaluation Gate + Runtime Monitoring Agent. Policy-as-code via .cognarc.yml. Soft-block and hard-block modes. All enforcement governed by the Trust Gradient. Human override always available.
Creative & Design Agent	Creative Cognitive Evaluator, Variant Ranker, Zero-Traffic A/B Engine, Onboarding Load Curve Agent, Prototype Heatmap Generator. All trigger automatically on upload or deployment detection. No submission workflow required.
Eval API	Custom cognitive scorer endpoint compatible with Braintrust, Langfuse, W&B Weave, and Arize Phoenix. Prompt Regression Cognitive Gate. Published OpenAPI 3.1 spec. Python and TypeScript SDK clients.
Benchmark & Audit Suite	Standardised cognitive assessment library derived from TRIBE's validated experimental paradigm base. Model cognitive profiles. Experience cognitive audits. Manipulation taxonomy reports. Regulatory compliance documents. All compiled autonomously, transmitted only with human approval.

How Humans Stay in Control

The oversight architecture was designed before any feature. Agentic systems that earn trust do so by making their governance explicit, not implicit. CognArc's Trust Gradient Engine is the structural proof that the agent's autonomy never exceeds the trust it has earned.

The Trust Gradient

Every possible CognArc action is classified into one of four oversight zones. Zone classification is determined by the consequence of the action, not the agent's convenience. The agent cannot reclassify its own actions to a lower oversight tier.

Agent Action	Zone	Human Role	How It Works
Monitor, score, and label outputs	Observe	**Reviews on demand**	Fully autonomous. Agent observes continuously. No threshold crossed. No action taken. Human reviews dashboard at any time.
Surface a cognitive risk flag	Observe	**Sees all flags**	Flag generated automatically. Agent surfaces in dashboard. Human decides whether to investigate.
Generate a rewrite recommendation	Recommend	**Approves or rejects**	Agent prepares evidence + ranked alternatives. Human makes the decision. Outcome recorded as a learning signal.
Post a PR comment with score breakdown	Act — Auto	**Reviews audit log**	Low-consequence notification. Logged. Reversible within 24hrs. Human can override at any time.
Soft-block an output (log + alert)	Act — Auto	**Reviews alert; can override**	Output flagged, not suppressed. Human can dismiss the soft block in real time.
Fail a CI/CD build on threshold breach	Act — Gated	**Configures thresholds**	Agent enforces human-configured policy. Override requires documented justification, logged to audit trail.
Hard-block a production output	Act — Gated	**Sets policy + can override**	Block mode and threshold set by human in .cognarc.yml. Override available in real time. Agent never modifies block policy.
Deploy an agent-generated prompt rewrite	Act — Gated	**Approves before deployment**	Agent generates and scores. Engineer approves, modifies, or rejects. Hard rule: never auto-deployed.
Execute a TRIBE fine-tuning run	Act — Gated	**Approves data + parameters**	Hard gate. No fine-tuning run executes without explicit human approval of training data and run configuration.

Agent Action	Zone	Human Role	How It Works
Transmit a regulatory audit report	Act — Gated	**Reviews, approves, signs**	Agent compiles. Human reviews, approves, and authorises. Agent never files regulatory documents autonomously.

The Three Structural Guarantees

- **One-click pause of all Act-auto and Act-gated agent actions across a workspace within 5 seconds. Passive monitoring continues. All queued actions cancelled. Cannot be disabled or circumvented by the agent under any circumstances.** Global Kill Switch:
- **Every agent action — autonomous or human-approved — written to an append-only audit log within 2 seconds. Log includes: timestamp, action type, oversight zone, triggering TRIBE scores, policy rule, alternatives considered, and authorising human or policy version. The agent has no write access to its own audit trail.** Immutable Audit Log:
- **The agent enforces .cognarc.yml policy. It does not write it, modify it, or reclassify its own actions against it. All policy changes require a human-authored commit with identity and timestamp. The agent rejects policy changes without a valid commit signature.** Policy Immutability:

> **Why human oversight is a commercial advantage, not a constraint:**
> *Enterprise buyers in fintech, healthcare, and regulated industries will not deploy an autonomous cognitive agent without demonstrable, auditable human control. CognArc's Trust Gradient — with logged, reversible, human-overridable autonomy at every consequence tier — is the feature that closes regulated-industry deals. Building oversight in from the start means it is architecturally sound, not bolted on. That is the difference a PM with governance foresight makes.*

Nine Pillars, One Always-On System

CognArc is structured around nine feature pillars. The first eight cover the product surface. The ninth — Human Oversight & Control — governs all eight. I will describe each briefly, with emphasis on the agentic design decision behind it and how human oversight is embedded.

A1 · Agentic Cognition Engine

TRIBE is CognArc's perceptual system, not a tool it calls. The engine ingests all connected AI outputs continuously, scores every stimulus in real time, and proactively generates counterfactual alternatives when risk is detected. Population-level variance modeling and cognitive heatmap generation both trigger automatically. The agentic decision: always-on continuous scoring, because cognitive risk doesn't wait for evaluation windows. Oversight: counterfactual deployment always requires human approval.

A2 · Autonomous Behavioral Intelligence

The behavioral layer is a persistent sensing system. Every SDK event is cognitive-labeled in real time. Analytics platform sync is live and bidirectional without scheduling. Alignment drift triggers automatic root cause analysis. The agentic decision: behavioral data is only valuable if it is current. Stale batch processing defeats the purpose. Oversight: TRIBE fine-tuning is the hardest-gated operation in the system — no run executes without explicit human approval.

A3 · Marketing & Growth Agent

Marketing teams receive cognitive evaluation the moment they upload an asset, not when they remember to request one. Variant ranking is automatic. Brand trust drift is monitored longitudinally without scheduling. The agentic decision: marketing teams will not use an evaluation tool on every asset. They will use an agent that evaluates every asset automatically. Oversight: agent ranks and recommends. Human makes every creative and deployment decision.

A4 · UI/UX Cognitive Testing Agent

Every onboarding flow update triggers an automatic cognitive load curve. Every Figma frame tagged for review receives a heatmap before the design review meeting. Comprehension gaps and trust timing mismatches are flagged continuously. The agentic decision: designers should arrive at user research with neuroscientific evidence, not open questions. Overnight evaluation after submission defeats that purpose. Oversight: agent surfaces all recommendations as supervised escalations. No UI changes made autonomously.

A5 · Manipulation & Trust Detection Agent

The manipulation scanner runs on every output that passes through CognArc — not a sample, not a scheduled batch, every output. Trust erosion is tracked longitudinally. Regulatory audit reports are compiled automatically on a schedule. The agentic decision: manipulation that affects 0.1% of outputs is still unacceptable at scale. Sampling doesn't catch it. Oversight: hard-block policy is always human-configured. Regulatory reports are never transmitted without human review and signature.

A6 · Autonomous Guardrail Enforcement

The guardrail layer self-operates: CI/CD evaluations trigger on PR detection, the Prompt Evaluation Gate intercepts submissions before they reach the LLM, the Runtime Monitoring Agent scores outputs at inference time. When a prompt fails, the agent generates a cognitively-optimised rewrite and presents it as a one-click replacement. The agentic decision: guardrails only work if they run on everything, automatically. A guardrail you have to remember to invoke isn't a guardrail. Oversight: rewrite deployment is always human-approved. Block policy is always human-configured.

A7 · AI Eval Platform Integration Agent

CognArc maintains live cognitive scoring connections to all connected eval platforms. Every output evaluated in Braintrust, Langfuse, W&B, or Arize is automatically scored in parallel. Prompt regression monitoring is continuous, not periodic. The agentic decision: cognitive scoring is most valuable when it appears alongside every other eval metric, automatically, without additional engineer steps. Oversight: platform connections require human approval. Cognitive scores are advisory within the eval platform context.

A8 · Benchmarking & Standards Agent

The benchmarking agent runs cognitive assessments on a configured schedule without instigation. Experience audits are triggered on connected product URLs automatically. Compliance reports are compiled on schedule. The agentic decision: a benchmark that runs when someone remembers to run it produces a point-in-time snapshot. A benchmark that runs continuously produces a cognitive health trend. Oversight: all external-facing outputs require human approval before distribution. Regulatory documents are never filed autonomously.

A9 · Human Oversight & Control Layer

This pillar does not have features in the conventional sense. It is the architecture. Trust Gradient Engine. Global Kill Switch. Immutable Audit Log. Policy-as-Code. Act-Gated Decision Package Workflow. Oversight Compliance Report. These are not add-ons to the other eight pillars — they are the structural layer without which none of the other eight should exist. The agentic decision: oversight must be designed first. Every feature must be built inside its constraints, not retrofitted to comply with them afterward.

How CognArc Syncs With Your Analytics Stack

CognArc does not replace your analytics platform. It sits one layer above it — giving behavioral data the cognitive interpretation it currently lacks. The integration architecture is built around one principle: CognArc should enrich the tools teams already use, not force them to abandon them.

The Integration Architecture

CognArc connects to analytics platforms via three mechanisms operating simultaneously.

- **CognArc subscribes to your existing behavioral event stream via Segment webhook, SDK middleware, or direct connector. Every event that flows through your analytics stack also flows into CognArc's cognitive labeling pipeline. No duplicate instrumentation required.** Event stream ingestion:
- **The agent labels each event with a cognitive state within 200ms of occurrence — using the distilled TRIBE model as the labeling engine. Rage click becomes confusion. Dwell-with-no-scroll becomes cognitive load stall. Field re-entry becomes working memory overload. These labels are appended to the event before it reaches your analytics platform's storage.** Real-time cognitive enrichment:
- **CognArc pushes cognitive scores back into your analytics platform as custom event properties. Your existing Amplitude or Mixpanel dashboards can now filter funnels, segment cohorts, and build charts by cognitive signal — without leaving the tools your team already knows.** Bidirectional write-back:

The Event-to-Cognition Translation Layer

The most important integration feature is one that requires no configuration: the Event-to-Cognition Translation Engine maps raw behavioral events to cognitive states automatically, based on TRIBE-validated signal patterns.

Raw Analytics Event	Cognitive Signal CognArc Labels	What It Means for the Product
Long dwell time + no scroll	Cognitive Load stall	User is stuck. The content is not being processed. Likely a comprehension failure.
Rage click	Confusion / violated expectation	User expected something and it did not happen. Interface contract is broken.
Form field re-entry (3+ times)	Working memory overload	Too many things to remember simultaneously. Decision architecture needs simplification.

Raw Analytics Event	Cognitive Signal CognArc Labels	What It Means for the Product
Drop-off at specific funnel step	Trust gap or decision fatigue	Either the product asked for something before earning it, or too many decisions accumulated.
High scroll velocity, low clicks	Low attention engagement	Content is being skipped. Comprehension Confidence is low. Value prop is not landing.
Repeated visits to same page	Unresolved comprehension	User returned because they did not understand on first visit. Copy or structure needs work.
Session abandonment post-modal	Trust erosion trigger	A specific interaction broke the trust relationship. Likely a premature ask or a dark pattern.

Practical Outcome for Product Teams

A PM using Amplitude can now filter their activation funnel by Cognitive Load level and see which steps are cognitively heavy — before looking at any behavioral drop-off data. They can segment their retention cohorts by Trust Coherence score and see whether low-trust users churn faster. They can build a dashboard that shows, in real time, whether a shipped change improved or degraded the cognitive experience.

The TRIBE simulation predicted that change's cognitive impact before it shipped. The behavioral data now confirms or challenges that prediction. The Cognitive-Behavioral Alignment Score surfaces the gap between the two — and when that gap is statistically significant, CognArc escalates it automatically for investigation.

> **The integration insight that shaped this feature:**
> *Behavioral analytics tells you what users did. CognArc tells you why they did it. Write-back closes the loop: the cognitive interpretation lives inside the behavioral tool, not in a separate dashboard that nobody checks.*

09 · MARKETING & GROWTH INTELLIGENCE

How Marketing Teams Use CognArc for Growth

Marketing operates at the intersection of cognition and commerce. Every campaign, every landing page, every piece of copy is an attempt to capture attention, build comprehension, generate trust, and convert. CognArc gives marketing teams a scientific instrument for what they have previously done by intuition and A/B testing — with the specific advantage that it works before media spend, not after.

Replacing Media Spend With Simulation

The most immediate value for a growth team: run five creative variants through CognArc before spending a dollar on media. TRIBE predicts which variant produces the highest attention capture above the fold, the strongest comprehension of the value proposition, the best emotional valence, and the most coherent trust signal. Ship the cognitive winner. A/B testing then validates the prediction rather than discovering it from scratch.

This is not a marginal improvement in test efficiency. For a team spending $50K/month on paid acquisition, running even one creative test pre-launch through CognArc rather than in-market can shift the economics of growth meaningfully. The cognitive simulation costs minutes. The in-market test costs weeks and budget.

Diagnosing the Real Cause of Low Conversion

Low conversion on a landing page has two possible causes: a targeting failure (wrong audience) or a comprehension failure (right audience, wrong message). Standard analytics cannot distinguish between them. CognArc can. The Message Clarity Scorer returns a Comprehension Confidence score for any piece of copy. If the score is low, the problem is the message, not the audience. If the score is high and conversion is still low, the problem is targeting. This distinction changes the optimization strategy entirely.

The Cognitive Funnel — A New GTM Primitive

CognArc introduces a concept that traditional funnel analytics cannot produce: the cognitive journey. Rather than showing where users drop off, it maps why — as a continuous sequence of cognitive states from first ad impression through to conversion. A growth team can see where cognitive load accumulates across the funnel, where trust first forms and where it breaks, and which touchpoints are doing emotional and cognitive work that downstream touchpoints are not.

Ad creative	Agent scores attention capture and emotional valence. Predicts whether the creative generates genuine curiosity or exploits urgency. Flags the latter as a manipulation risk.

Landing page	Cognitive Audit scores all four dimensions. Identifies whether the value proposition is understood within 3 seconds of above-the-fold exposure. Flags comprehension stalls.
Sign-up flow	Onboarding Load Curve shows where friction accumulates. Trust Timing Analyzer flags steps asking for personal data before trust has been established.
Onboarding sequence	Per-step cognitive scores identify where users are most likely to drop off and why. Choice overload, comprehension gaps, and trust timing mismatches all surfaced before launch.
Retention emails	Message Clarity Scorer and Manipulation Risk Scorer applied to every batch. Brand trust drift monitored longitudinally. Re-engagement campaigns scored before send.

Brand Trust as a Growth Metric

The most underused growth metric is longitudinal trust. Trust Coherence scores tracked across a user's sessions reveal whether repeated AI and product interactions are building trust or degrading it. A brand that slowly erodes trust through minor manipulative patterns — artificial urgency, social proof fabrication, dark patterns in pricing — will see churn before it understands the cause. CognArc surfaces the erosion signal weeks before behavioral data confirms it.

> **The growth team use case in one sentence:**
> *CognArc turns the marketing funnel from a behavioral measurement into a cognitive diagnostic — and moves the optimization decision from after spend to before it.*

A/B Testing Without Traffic — Startups & Lean Teams

Traditional A/B testing has a fundamental constraint that most product teams accept as permanent: you need traffic to reach statistical significance. A startup with 500 monthly visitors cannot run a valid test on two landing page variants in any meaningful timeframe. They either ship blind, copy what worked for someone else, or spend months waiting for a sample size.

CognArc breaks that constraint. The A/B test runs in simulation before a single real user sees either variant. TRIBE generates predicted brain responses for both. The cognitive winner is identified. The team ships it. Behavioral instrumentation then validates the prediction as real traffic arrives — rather than discovering the answer from scratch.

How It Works for a Landing Page

Step 1 — Upload	Submit two landing page variants as PNG, JPG, or rendered HTML. The agent triggers automatically on upload. No configuration required.
Step 2 — Simulate	TRIBE runs parallel inference on both variants. Predicts brain responses across attention, cognitive load, comprehension, emotional valence, and trust coherence for each.
Step 3 — Compare	Comparative cognitive report generated within 5 minutes. Per-variant scores across all four dimensions. A recommended winner with a plain-language rationale. Shareable link valid 30 days.
Step 4 — Ship	Team ships the cognitive winner. Behavioral SDK begins collecting real interaction signals from actual visitors.
Step 5 — Validate	As traffic arrives, CognArc computes the Cognitive-Behavioral Alignment Score: how closely real behavior matches TRIBE's prediction. Divergence triggers investigation.
Step 6 — Improve	Validated data refines future simulations. The agent gets more accurate at predicting how your specific audience responds over time.

Onboarding Flow Cognitive Analysis

Onboarding is where cognitive load kills retention. Most teams discover this in churned user interviews, weeks after the damage is done. CognArc surfaces it before launch — automatically, on every flow update.

The Onboarding Load Curve Agent evaluates each step of a flow and generates a cognitive load profile across the entire sequence. It shows exactly where demand peaks, where comprehension drops, and where the product asks for trust before it has been earned. Three specific detectors run on every evaluated flow:

- **Flags steps where predicted Comprehension Confidence falls below the workspace threshold — identifying copy that assumes knowledge the user does not have. The primary driver of step-level abandonment that analytics misattributes to friction.** Comprehension Gap Detector:
- **Detects when a flow asks for trust — a permission request, a payment step, a personal data entry — at a moment where TRIBE's Trust Coherence prediction is below 50. The product is asking for something it has not yet earned.** Trust Timing Analyzer:
- **Flags steps where TRIBE's prefrontal load signature indicates simultaneous decision count exceeds working memory capacity. The cognitive version of decision fatigue — invisible in analytics until dropout has already occurred.** Choice Overload Detector:

The Startup Advantage

For a seed-stage founder or a lean product team, CognArc compresses the feedback loop from weeks to hours and makes rigorous product decisions accessible without a research team. You arrive at user research sessions with neuroscientific hypotheses rather than open questions. You ship the onboarding flow with cognitive evidence rather than a best guess. You know which variant is cognitively stronger before you have traffic to prove it statistically.

> **The constraint CognArc removes for early-stage teams:**
> *Statistical significance requires sample size. Sample size requires traffic. Traffic requires time. CognArc replaces that dependency chain with cognitive simulation — making the quality of your product decisions independent of the size of your user base.*

CognArc in the AI Eval Ecosystem

CognArc is not a competitor to existing AI evaluation platforms. It is the cognitive layer that makes them more meaningful. The integration strategy is built around one strategic decision: own what no one else can build, and integrate cleanly with everything that already exists.

The Strategic Integration Decision

Existing AI eval platforms — Braintrust, Langfuse, Weights & Biases, Arize — are good at what they do: measuring task performance, hallucination rates, faithfulness, and human preference. These are necessary measures. They answer whether the AI got the answer right. They do not answer what that answer does to the person receiving it. CognArc answers the second question and connects it to the first.

Capability	Decision	Rationale
Hallucination detection	Integrate	Well-solved by Galileo, Ragas, TruLens. CognArc layers cognitive trust scoring on top — a hallucination that erodes trust scores higher risk than one that does not.
Human preference scoring	Integrate	Complements cognitive scoring. CognArc distinguishes whether a preferred output is preferred because it is genuinely clear or because it is persuasive.
Behavioral analytics collection	Integrate	Amplitude, Mixpanel, PostHog collect. CognArc interprets and writes back cognitive labels.
Toxicity classification	Integrate	CognArc adds the manipulation and cognitive confusion layer above existing toxicity classifiers.
Cognitive impact scoring	Own	CognArc's primary moat via TRIBE v2. No equivalent exists in the market. This capability is not integrable — it must be built.
Dark pattern detection	Own	Existing tools are rule-based. CognArc adds brain-response grounding and neural evidence to every detection.
Prompt regression testing	Own	CognArc's CI/CD gate catches cognitive regressions that accuracy-only tests miss. A prompt that passes accuracy checks but increases user confusion by 20 points is a regression.
Benchmark leaderboards	Own	CognArc's benchmark suite is differentiated by TRIBE's neuroscientific foundation. Not replicable by task-accuracy metrics.

How the Integration Works Technically

CognArc exposes a Cognitive Scorer API that conforms to the custom scorer interface used by all major eval platforms. Input: a model output string plus optional context. Output: a structured JSON score object containing Cognitive Load, Comprehension Confidence, Emotional Valence, Trust Coherence, and Manipulation Risk, each with a confidence interval and a plain-language explanation.

Any eval platform that supports custom scorers — which Braintrust, Langfuse, W&B Weave, and Arize Phoenix all do — can call this endpoint and display cognitive scores alongside accuracy, faithfulness, and toxicity scores in their existing dashboard. For engineering teams already running evals, the integration requires no new tooling and no new dashboard. Cognitive scores appear in the tool they already use.

The Prompt Regression Cognitive Gate

The most immediately valuable integration for an engineering team is the Prompt Regression Cognitive Gate. It works alongside existing accuracy regression tests in CI/CD. When a new model version or prompt variant is evaluated, CognArc compares cognitive scores against a stored baseline. A regression is defined as: Cognitive Load increase greater than 10 points, or Comprehension Confidence decrease greater than 15 points versus baseline.

This catches a failure mode that accuracy-only regression tests routinely miss: a prompt change that produces equally accurate outputs but makes them significantly harder for users to understand. In a consumer AI product, that is a regression that matters — it just does not show up in any standard eval metric.

What to Integrate vs. What to Absorb

The integration decision matrix above reflects a clear product philosophy: CognArc does not rebuild what already exists and works. It builds what does not exist and cannot be replicated without TRIBE. The Cognitive Scorer API is designed to make CognArc the standard cognitive evaluation layer that every AI eval platform integrates into — not another tool requiring another dashboard login.

CognArc as a Red Team Force Multiplier

Red teaming is one of the most important practices in AI safety — and one of the most structurally limited. Human red teams are skilled, but they are periodic, they evaluate what they think to probe, and they cannot scale to the volume of outputs a modern AI system produces. CognArc does not replace red teaming. It makes red team findings more credible, more durable, and vastly more complete.

What Red Teams Currently Lack

- **A human red team evaluating hundreds or thousands of outputs will never encounter the edge cases that only appear at millions of outputs per day. The 0.1% of outputs with a manipulation signature, the subtle trust-eroding pattern that emerges only under specific conversational conditions — these are the failures that cause reputational and regulatory damage, and they are precisely the ones that periodic manual red-teaming misses.** Coverage at scale:
- **When a red teamer identifies a manipulative output pattern, they produce a finding based on judgment and experience. That finding is an opinion. It can be disputed, deprioritized, or overridden by stakeholders who weigh it against product velocity. A TRIBE-grounded cognitive evidence package — showing the limbic activation signature, the prefrontal engagement ratio, and the neural mechanism of harm — transforms a finding from an opinion into a scientifically defensible claim.** Scientific grounding for findings:
- **Red teams fix something and move on. They have no continuous mechanism for verifying that the fix held. A model update three weeks later may re-introduce the same manipulation pattern they identified and remediated. Without monitoring, that regression is invisible until the next red team cycle.** Regression monitoring post-remediation:
- **Red team findings need to survive regulatory scrutiny. An immutable, timestamped audit log of every manipulation flag, every block event, and every human remediation decision is exactly what a safety team needs when preparing evidence for an internal governance review, a board report, or an external regulatory audit.** Regulatory-grade documentation:

How CognArc Augments a Red Team Function

Red Team Need	Current Gap	CognArc Capability
Continuous manipulation coverage	Periodic manual testing only	Continuous manipulation scanner across all 6 taxonomy categories, running on every output, 24/7.
Neural evidence for findings	Opinion-based findings, easily deprioritized	TRIBE-grounded cognitive evidence: brain region activation, limbic-prefrontal ratio, confidence interval.

Red Team Need	Current Gap	CognArc Capability
Scale beyond human capacity	Hundreds of test cases per cycle	Every output evaluated — millions per day if needed. Edge cases at scale become visible.
Post-remediation regression watch	No monitoring after a fix is shipped	Autonomous Prompt Regression Monitor detects re-emergence of flagged patterns the moment a model update is connected.
Longitudinal trust erosion signal	Only visible in user research, weeks after	Trust Coherence tracked per session longitudinally. Slow-burn erosion surfaced within 7 days.
Regulatory-grade audit trail	Red team reports are documents, not evidence logs	Immutable audit log of every flag, every block, every human decision. Timestamped, attributable, exportable.
Population-level harm detection	Aggregate analysis hides subgroup risk	Population-Level Variance Engine continuously flags outputs harmful to specific demographic cohorts.

The Positioning for AI Safety Teams

CognArc is not a red teaming tool. Red teaming requires adversarial creativity, domain expertise, and human judgment that no agent can replicate. What CognArc provides is the continuous monitoring layer that makes red team findings stick: it catches regressions after a fix ships, it surfaces manipulation patterns at a scale human red teams cannot reach, and it produces the scientific and evidentiary infrastructure that turns a safety team's findings into defensible organizational and regulatory claims.

For an AI safety team operating under regulatory scrutiny — any company subject to the EU AI Act, any frontier lab with a safety commitment, any enterprise deploying AI in healthcare, finance, or education — CognArc is the infrastructure that makes the safety posture auditable, not just asserted.

The red team positioning in one sentence:

Red teams find what to fix. CognArc ensures the fix holds, catches what red teams cannot see at scale, and produces the evidence trail that makes safety claims credible to regulators and boards.

How CognArc Reaches the Market

The GTM Insight

Every buyer CognArc serves carries a version of the same quiet frustration: they make decisions about cognitive quality — in a prompt, a campaign, a flow, a model update — with no instrument to measure it. The engineer ships a prompt change hoping it didn't make outputs harder to understand. The growth team launches a campaign hoping the copy was actually comprehended. The designer publishes an onboarding flow hoping users won't drop off at step three. The red team ships a fix hoping the pattern doesn't re-emerge. Hope is the current state of the art.

CognArc is not sold as an evaluation tool with agentic features. It is sold as the end of that. The GTM message is built around a concrete before-and-after: before CognArc, you discover cognitive problems when users complain. After CognArc, you know before they ever see it.

Lead Motion: Developer-Led Growth via Free Agentic Hook

The initial motion is developer-led. The free-tier hook is the Autonomous Prompt Regression Monitor: connect any LLM endpoint in five minutes, receive continuous cognitive regression alerts without scheduling an eval run. Engineers feel the value within the first session. No dashboard required, no procurement cycle, no sales call. The product sells itself the first time it catches a regression the engineer would have missed.

Developer adoption creates three assets that unlock every subsequent motion: behavioral data that improves TRIBE simulation accuracy over time, organisational champions inside accounts who become internal advocates for expansion, and the usage credibility that makes the enterprise and regulatory sales motions faster to close.

Secondary Motion: Product-Led Expansion Within Accounts

Once an engineering team is active, the expansion motion runs product-led. The CI/CD gate surfaces cognitive scores in pull requests. PMs start asking what those scores mean for onboarding. Growth teams ask whether the same scoring applies to landing pages. Designers ask whether it works on Figma prototypes. CognArc is designed so that each of those questions has an immediate answer and a frictionless entry point — the product expands horizontally within accounts without a sales conversation.

The five-buyer architecture is an expansion strategy, not just a segmentation model. Every new buyer surface activated within an account increases connection depth, which increases ACV, which increases retention. Expansion is the natural consequence of the product working, not a separate GTM motion.

Segment Prioritization

The sequencing below reflects both buyer readiness and strategic value. Engineers move fastest and create the account foothold. Growth and design teams expand it. Safety and compliance teams lock it in. Regulators and auditors validate it publicly.

Tier & Segment	Core Problem	Primary Hook	Expansion & Strategic Value
Tier 1 · AI Product Teams	Cognitive regressions invisible to accuracy-only eval. Manual eval burden doesn't scale.	Autonomous Prompt Regression Monitor (free). CI/CD Cognitive Gate. Prompt Evaluation Gate.	Creates the account foothold. Developer adoption generates behavioral data that improves TRIBE. Highest velocity, lowest CAC. Expands to PM and design surfaces within 90 days of activation.
Tier 2 · Growth & Marketing Teams	No cognitive instrument for creative testing before media spend. Trust erosion invisible until conversion drops.	Creative Cognitive Evaluator. Variant Ranker. Brand Trust Drift Monitor.	High-budget buyers with measurable ROI: cognitive simulation replacing media spend on failed creative. Often the team that drives Growth-tier conversion within an account. Strong word-of-mouth surface.
Tier 3 · Design & Product Teams	No cognitive evidence for design decisions before user research. Startup constraint: no traffic for valid A/B tests.	Zero-Traffic A/B Engine. Onboarding Load Curve Agent. Prototype Cognitive Heatmap.	High engagement, high stickiness. Designers who use CognArc before every user research session become internal champions. Particularly strong in startups and scale-ups where research resources are thin.
Tier 4 · AI Safety & Red Teams	Manual, periodic red-teaming at a fraction of output volume. Findings unverified post-fix. No continuous coverage.	Continuous Manipulation Scanner. Trust Erosion Monitor. Immutable Audit Log.	Creates the compliance and governance infrastructure that locks in Enterprise contracts. The safety team's adoption of CognArc is the event that triggers procurement by legal and compliance — the highest-ACV motion in the account.
Tier 5 · Regulators & Auditors	No independent, reproducible cognitive assessment standard exists. Regulatory claims about AI safety are asserted, not evidenced.	Cognitive Benchmark Suite. Signed Audit Reports. Regulatory Compliance Documentation.	Longest sales cycle, highest ACV, strongest market signal. A government body or independent auditor adopting CognArc's benchmark suite as a standard creates a category-defining moment. Sequenced last because it requires the credibility built by Tiers 1–4.

Pricing Architecture

Pricing is structured on connection depth — the number of AI systems, analytics platforms, and buyer surfaces connected — not evaluation volume. A connected system generates continuous value regardless of dashboard activity. Deeper integration drives higher ACV, higher retention, and natural expansion revenue without a separate upsell motion.

Free · Developer	Up to 3 connected endpoints. Continuous cognitive scoring. Autonomous Prompt Regression Monitor. Audit log (30 days). SDK + 1 analytics connector. The hook that creates the account foothold.
Growth · $799/mo	Unlimited connected endpoints. Full Trust Gradient Engine. CI/CD gate. Prompt Evaluation Gate. All analytics connectors. Continuous manipulation scanner. Creative evaluator. Zero-traffic A/B engine. 90-day audit log.
Business · $3,499/mo	All Growth. Runtime Monitoring Agent. Autonomous prompt remediation. Self-healing incident response. Trust erosion monitor. Onboarding load curve agent. Regulatory audit reports. SLA. The tier that activates safety and compliance buyers.
Enterprise · Custom	All Business. TRIBE fine-tuning pipeline (human-gated). On-prem deployment. Custom oversight compliance reporting. Dedicated advisory support. Researcher API access. The tier that closes regulated-industry and government contracts.

Infrastructure Design and Rationale

The infrastructure architecture for CognArc is shaped by three non-negotiable operating requirements: continuous inference at sub-600ms latency, event-driven agent operation at production scale, and human-gated governance for every consequential operation. The decisions below flow from those requirements.

Decision 1: Two-Tier TRIBE Inference

Continuous inference on a full research-grade foundation model is not feasible at production latency. CognArc uses a distilled TRIBE model for all synchronous continuous scoring — trained to approximate the full model's predictions at a fraction of the compute cost, targeting p95 <300ms on the inference layer. The full model runs on async GPU infrastructure for operations that can tolerate minutes: counterfactual generation, population variance modeling, benchmark suite runs, and regulatory audits. Stimulus hash caching prevents redundant inference on repeated outputs, critical when continuous scoring can process millions of outputs per day.

Decision 2: Event-Driven Agent Pipeline

The agent is event-driven rather than polling-based. Every connected AI system and analytics platform emits events to CognArc's ingestion layer via webhook or SDK. The agent processes events in a real-time stream — Apache Kafka for ingestion, distilled TRIBE for cognitive labeling, ClickHouse for columnar analytics storage. Anomalies are routed to the appropriate action queue based on Trust Gradient zone classification. The Trust Gradient router is the architectural backbone: every event that crosses a detection threshold is classified before any action executes.

Decision 3: WORM Audit Storage

The audit log is stored in write-once read-many (WORM) storage, replicated to a separate system within 30 seconds of entry creation. The agent has no write access to audit storage. This is not a security feature — it is a governance feature. The audit log is the evidentiary foundation of every regulatory claim CognArc makes. Its integrity must be structurally guaranteed, not administratively managed.

Decision 4: GitOps-Based Policy Management

Policy configuration (.cognarc.yml) is managed as code in the customer's repository. Changes require a human-authored commit with identity and timestamp. The agent validates commit signatures before applying policy changes. This means every governance decision is versioned, attributable, and auditable alongside the code it governs. The agent cannot modify policy by any mechanism — not through the API, not through the dashboard, not through any action it is capable of taking.

Three Phases, Five Buyer Motions

The roadmap is sequenced by three constraints operating simultaneously: technical risk (continuous inference before fine-tuning), oversight maturity (Trust Gradient architecture must be complete and stable before expanding agent autonomy in Phase 2), and buyer motion (each phase must deliver a complete, independently valuable experience for at least one buyer type — not a partial feature set for everyone).

The sequencing is deliberate. Phase 1 activates the two buyer types with the shortest sales cycles and the highest evaluation velocity: engineers and PMs. Phase 2 activates growth, design, and safety buyers — who require the behavioral validation layer and deeper agentic capabilities to see their core value. Phase 3 builds the compounding infrastructure that creates the long-term moat and unlocks the regulatory and academic market.

Phase 1 · Foundation (Months 0–8)

The agentic core ships complete. The oversight architecture — Trust Gradient Engine, Global Kill Switch, immutable audit log — is live from day one. No agentic capability ships without its governance layer already in place.

What ships	Continuous TRIBE inference. Autonomous Prompt Regression Monitor. CI/CD Cognitive Gate. Prompt Evaluation Gate. Persistent Behavioral SDK. Analytics connectors (Segment, Amplitude, Mixpanel, PostHog, GA4) with write-back. Autonomous Manipulation Scanner. Creative Cognitive Evaluator. Variant Ranker. Zero-Traffic A/B Engine. Onboarding Load Curve Agent. Cognitive Scorer API (Braintrust, Langfuse, W&B, Arize).
Buyer motion live	AI Engineers: CI/CD gate and prompt regression monitor operational. PMs: alignment scoring and analytics cognitive enrichment live. Growth teams: creative evaluator and variant ranker available. Designers: zero-traffic A/B and onboarding load curve operational.
Definition of done	First 100 paying workspaces. Free-to-Growth conversion rate > 12%. False positive rate on triggered alerts < 8%. TRIBE distilled model Pearson r > 0.72. SOC 2 Type I in progress.
Phase 1 milestone	The product is useful to four of five buyer types on day one. The engineering and PM buyers have enough to generate champions. Growth and design buyers have enough to convert.

Phase 2 · Scale (Months 9–18)

Expand agentic depth and activate the safety and compliance buyer motion. Phase 2 is where CognArc moves from a tool individual teams use to a platform organisations depend on.

What ships	Runtime Monitoring Agent (hard-block). Autonomous prompt remediation. Self-healing incident response. Real-Time Confusion Blocker. Dark Pattern Agent. Trust Erosion Monitor. Session replay cognitive annotation. Proactive counterfactual generation. Full tri-modal TRIBE inference (audio + video). Eval platform native connectors. Cognitive Benchmark Suite. Experience Cognitive Audit. Regulatory Audit Reports. Oversight Compliance Report.
Buyer motion live	AI Safety & Red Teams: continuous manipulation scanner at full depth, trust erosion longitudinal tracking, and regulatory-grade audit trail all operational. Regulators & Auditors: benchmark suite and signed audit reports available. All five buyer surfaces now fully active.
Definition of done	500 active workspaces. First Enterprise contracts closed. Net Revenue Retention > 115%. Runtime Monitoring Agent deployed in at least 20 production environments. Oversight Compliance Report generating for all Business-tier accounts.
Phase 2 milestone	The platform is sticky across all five buyer types. The safety team's adoption triggers Enterprise procurement. The regulatory motion begins.

Phase 3 · Platform (Months 19–30)

Close the compounding loop. Build the infrastructure that creates network effects and category ownership. Phase 3 is where CognArc stops being a product and becomes a standard.

What ships	TRIBE fine-tuning data pipeline (human-gated at every stage — the hardest governance gate in the system). Longitudinal Benchmark Regression Monitor. Researcher API. Public Cognitive Leaderboard. Segment-Level Message Optimizer. Human Preference + Cognitive Alignment Fusion.
Buyer motion live	Regulators & Auditors: public leaderboard creates a market incentive structure around cognitive alignment scores. Researcher API opens the benchmark suite to academic validation. Fine-tuning loop begins compounding TRIBE accuracy on accumulated behavioral data.
Definition of done	2,000 paying workspaces. $36K blended ACV. First government or regulatory body adopting CognArc benchmark suite as a reference standard. TRIBE fine-tuning loop producing measurable accuracy improvement vs Phase 1 baseline.
Phase 3 milestone	Network effects active. The benchmark suite is cited externally. The fine-tuning loop is compounding. CognArc is not a product competing in a category — it is the category.

The Commercial Case

The Revenue Architecture

CognArc prices on connection depth — the number of AI systems, analytics platforms, creative tools, design workflows, and safety functions connected to the agent — not on evaluation volume. This is a deliberate structural choice. A connected system generates continuous value regardless of whether anyone is actively using the dashboard. The more surfaces a team connects, the more indispensable the agent becomes, and the higher the switching cost.

The account expansion model follows the GTM sequencing directly. An engineering team connects on the free tier. Their CI/CD gate and prompt regression monitor generate value immediately. A PM notices the cognitive scores in pull requests and asks for dashboard access. That converts the account to Growth. A growth team asks about the creative evaluator. A designer asks about the A/B engine. A safety lead asks about the manipulation scanner and compliance reports. Each question is a natural expansion trigger, not an upsell conversation. By the time a legal team gets involved for Enterprise procurement, the account has already proven value across three or four buyer surfaces.

Unit Economics

$36K	15mo	130%+	<$4K
Target Blended ACV (Yr 2)	Target Payback Period	Net Revenue Retention Target	Target CAC (Developer Motion)

The developer-led free tier keeps CAC structurally low. The five-buyer expansion model keeps NRR structurally high — an account that started as a single engineering team and expanded to cover growth, design, and safety functions has a very high cost of switching, independent of any lock-in mechanism. Value delivered at depth is the retention strategy.

Market Sizing

CognArc operates across five distinct markets simultaneously. The addressable market is the intersection of all five, not any single one.

AI Evaluation Tooling	$2.1B TAM (2025), growing 34% YoY. CognArc is the only cognitive impact layer in this market. Every AI eval platform is a potential integration partner and distribution channel.

Digital Experience Analytics	$12.8B TAM. Behavioral analytics platforms measure what users did. CognArc adds the cognitive interpretation layer that explains why. Write-back integration makes this an expansion surface, not a competitive one.
Creative & Marketing Intelligence	$4.2B TAM. Marketing teams spend billions testing creative after the fact. CognArc's pre-spend cognitive simulation is a new category within this market — no direct equivalent exists.
AI Safety & Compliance	Regulatory-driven. EU AI Act enforcement begins 2026. Estimated $4B+ by 2028. No cognitively-grounded tool currently exists. CognArc is positioned to define the standard before it is mandated.
Design & UX Research Tooling	$1.8B TAM. Design teams pay for user research to answer questions CognArc can simulate. The zero-traffic A/B engine and cognitive heatmap position CognArc as a research-acceleration tool, not a research replacement.
Serviceable Market (3yr)	$1.2B–$1.8B across all five surfaces. Target: 2,000 paying teams by end of Year 3 at $36K blended ACV. Growth from account expansion, not primarily new logo acquisition.

Why the Moat Compounds

Most SaaS moats are built on switching costs or network effects. CognArc has both, plus a third: simulation accuracy that improves with usage. Every validated behavioral session adds to the TRIBE fine-tuning dataset. Every account that uses CognArc makes the simulation more accurate for every account that follows. The platform gets better as it scales — not just more feature-rich, but more scientifically accurate. A competitor starting from scratch does not just face a feature gap. They face an accuracy gap that widens every month CognArc operates.

What Could Go Wrong

A product that spans five buyer types, three architectural layers, a research-grade foundation model, and an agentic governance system has a risk profile that crosses technical, commercial, and regulatory dimensions simultaneously. The risks below are organised by category. Each includes the phase in which it is most acute, because timing changes both the severity and the mitigation approach.

Technical & Scientific Risks

Distilled TRIBE accuracy insufficient for continuous use · Phase 1 critical	Distillation trades accuracy for inference speed. If the distilled model's Pearson r vs. the full model drops below 0.68 on digital stimuli, every cognitive score the platform produces loses credibility. Mitigation: hold-out validation set of digital stimuli pre-launch. NFR r > 0.72 monitored on a rolling basis. Full TRIBE available as async fallback. Domain-shift validation study completed before Phase 1 ship date, not carried as a known risk.
TRIBE fine-tuning degrades model accuracy · Phase 3 critical	The feedback loop is the most powerful and most dangerous mechanism in the system. It modifies the agent's own perceptual accuracy. If it runs on insufficient or low-quality data, it degrades the scientific foundation everything else is built on. Mitigation: hard human gate at every fine-tuning stage. Post-run validation against benchmark suite before any production promotion. Rollback mechanism. Phase 3 placement ensures minimum 18 months of behavioral data accumulation before activation.
False positive rate too high for agentic operation · Phase 1–2 critical	Continuous scanning at scale creates volume risk. A false positive rate above 8% makes the agent more friction than value — teams start ignoring alerts, which defeats the entire architecture. Mitigation: conservative default thresholds (soft-block only in Phase 1). False positive rate monitored as a primary KPI alongside ACV and NRR. Threshold tuning wizard in onboarding. Override feedback loop used to recalibrate agent recommendations over time.

Agentic Governance Risks

Trust gradient misconfiguration · Phase 1–2	Overly permissive policy gives the agent excessive autonomy and creates liability. Overly restrictive policy makes it useless and drives churn. Either misconfiguration is a product failure. Mitigation: policy templates per workspace type (dev / staging / prod). Policy change requires peer review commit. Weekly Oversight Compliance Report surfaces whether the configured trust gradient is actually working.

Agent hard-blocks production output incorrectly · Phase 2 critical	A hard-block error in a production environment is a customer-facing incident with potential SLA, legal, and reputational consequences. Mitigation: hard-block mode requires explicit human policy configuration. Override available via API within 30 seconds. Default threshold set conservatively. Hard-block capability withheld to Phase 2, after Phase 1 establishes calibration data and customer trust.
Regulatory challenge to autonomous manipulation blocking · Phase 2+	Autonomous blocking of AI outputs could be construed as editorial censorship or content moderation under EU AI Act Article 9 in some jurisdictions. Mitigation: all autonomous blocks scoped to internal pipeline operations only. No autonomous action visible to end users without human review. Regulatory counsel retained before Phase 2 expansion. All block rationale logged to immutable audit trail.

Commercial & GTM Risks

Developer adoption doesn't convert to multi-buyer expansion · Phase 1–2	The five-buyer expansion model only works if free-tier engineering adoption creates enough value and visibility that adjacent teams ask to join. If engineers use CognArc in isolation and don't advocate internally, the account stays on the free tier indefinitely. Mitigation: product design actively surfaces cognitive scores in shared artifacts (PR comments, dashboards, reports) that PM, design, and growth teams see. The expansion trigger must be built into the product experience, not left to sales.
Growth and design buyers don't see ROI fast enough · Phase 1	Growth and design buyers need to feel value in the first session. If the creative evaluator takes 10 minutes and returns a score with no clear action, they will not return. Mitigation: time-to-value target of under 3 minutes for first cognitive report. Every score accompanied by a plain-language explanation and a ranked recommendation. Onboarding flow designed around the first successful simulation, not feature discovery.
Regulatory sales cycle longer than anticipated · Phase 2–3	Government bodies and independent auditors move on procurement timelines measured in years, not months. If the regulatory motion is relied upon for Phase 3 revenue targets, a single procurement delay creates a significant shortfall. Mitigation: regulatory buyers are Tier 5 in the GTM sequence — revenue model is not dependent on their conversion in Phase 3. They are a market signal and a moat builder, not a near-term revenue driver.
Competitor ships cognitive eval feature before CognArc scales · Phase 1–2	A well-resourced competitor (Braintrust, Langfuse, or a frontier lab's internal tooling team) could ship a cognitive scoring feature that reduces CognArc's differentiation. Mitigation: the moat is not the scoring feature — it is TRIBE's neuroscientific grounding, the accumulated behavioral fine-tuning data, and the five-buyer agentic architecture. A competitor shipping a heuristic-based cognitive scorer does not replicate any of those three. File defensive IP on the TRIBE productization layer. Move fast on the fine-tuning loop to widen the accuracy gap.

The Decisions Behind This Product

Every product is a series of choices about what to prioritise, what to defer, and what to give up entirely. This section documents the most consequential decisions behind CognArc — architectural, commercial, and strategic — including what I got right, what I would do differently, and what this body of work is designed to demonstrate.

Decision 1: Governance Before Features

The first design work I did on CognArc was not a feature list. It was a consequence matrix: for every possible action the agent could take, what is the worst realistic outcome if that action is wrong? That exercise produced the Trust Gradient before a single feature was scoped. Every pillar was then designed inside the governance constraints, not retrofitted to comply afterward.

This sequence matters for a reason that goes beyond safety. Governance that is designed first is architecturally coherent — every component knows its oversight zone, every action knows its escalation path. Governance that is bolted on afterward is inconsistent, full of edge cases, and unconvincing to the enterprise buyers and regulators who will scrutinise it most carefully. The decision to start with the Trust Gradient was a product quality decision, not just a safety one.

Decision 2: Five Buyers, Not Two

The default framing for a product like CognArc is two buyer types: Builders (engineers, PMs) and Evaluators (safety teams, regulators). That framing is clean and simple. It is also wrong for this product.

Growth teams, designers, and red teamers each have a distinct cognitive measurement problem that is not covered by either bucket. Collapsing them into Builders or Evaluators would have produced a product that was general enough to be marketed to everyone and specific enough to be genuinely useful to no one. The decision to design distinct value propositions, distinct entry points, and distinct feature surfaces for five buyer types added significant design complexity. It also produced a product with a natural account expansion model, a wider TAM, and a much stronger case for enterprise adoption. The complexity was worth it.

Decision 3: The Fine-Tuning Loop Goes in Phase 3

The TRIBE fine-tuning feedback loop is the capability that makes CognArc's simulation accuracy compound over time. It is the long-term moat. It is also in Phase 3, behind the hardest human gate in the system. I could have pushed for Phase 2 and shipped it earlier.

I chose not to because the fine-tuning loop modifies the agent's own perceptual accuracy. Activating it before sufficient behavioral data has accumulated, or before the Oversight Compliance Report has demonstrated a stable and well-calibrated agent, risks degrading the scientific foundation the entire platform is built on. Speed of capability accumulation is less important than trust in the agent's judgment. You cannot build trust faster than you demonstrate it. The Phase 3 placement is not timidity. It is the correct engineering and governance decision for an operation that consequential.

Decision 4: Integration Over Replacement

CognArc could have been positioned as a replacement for existing AI eval platforms, analytics tools, and design research services. That positioning would have been simpler to communicate and harder to sell. Every replacement product faces adoption resistance from the tools it is trying to displace.

The integration-first positioning — CognArc as the cognitive layer that makes existing tools more meaningful, not the tool that replaces them — is harder to explain but easier to adopt. Engineers don't need to leave Braintrust or Langfuse. PMs don't need to abandon Amplitude. Growth teams don't need a new dashboard. Cognitive scores appear inside the tools they already use. The decision to own cognitive impact scoring and integrate cleanly with everything else shapes the GTM motion, the pricing model, the API design, and the competitive positioning simultaneously. It is the single most commercially consequential architectural decision in the product.

What I Would Do Differently

I would have front-loaded the domain-shift validation study. TRIBE v2 was trained on naturalistic stimuli. Digital UI artifacts and LLM outputs are domain-shifted inputs. The credibility of every cognitive score CognArc produces depends on a rigorous empirical assessment of how well TRIBE generalises to those inputs. I documented this as a known risk rather than scheduling it as a pre-launch deliverable. That was the wrong call. It should have been the first thing on the engineering and research roadmap.

I would also have designed the free-tier onboarding flow alongside the product architecture, not after it. The five-buyer expansion model only compounds if each buyer type reaches a moment of genuine value in their first session. Designing that experience is as important as designing the underlying capability — and it requires the same level of product thinking. I treated it as a growth problem to solve later rather than a product design problem to solve now.

What This Work Demonstrates

This case study is a record of how a PM thinks at the architecture level — across technical, commercial, governance, and scientific dimensions simultaneously. The competencies it demonstrates:

Agentic system design:

Nine interdependent pillars, a four-zone trust gradient, and a compounding fine-tuning loop. Every autonomy boundary documented and justified from first principles, not convention.

Governance-first thinking:

Oversight architecture designed before any feature. The Trust Gradient, Kill Switch, and Audit Log are architectural artifacts with commercial value, not compliance add-ons added at the end.

Multi-buyer product strategy:

Five distinct buyer problems solved by one platform with one architecture. Distinct entry points, distinct value propositions, a natural account expansion model, and a TAM that spans five adjacent markets.

Scientific fluency:

Identified TRIBE v2 as the enabling research, understood its productization constraints, designed a path that exploits its generalization capability while mitigating its domain-shift risk — and placed the fine-tuning loop in Phase 3 for the right scientific reasons.

Integration and ecosystem thinking:

Explicit integrate vs. own decisions for eight overlapping capabilities. Positioned CognArc as the cognitive layer that makes existing tools more valuable — the decision that determines whether a product gets adopted or faces adoption resistance.

Commercial clarity:

Pricing on connection depth, not evaluation volume. Account expansion as the revenue model. A moat built on three compounding advantages: switching costs, network effects, and simulation accuracy that widens with every passing month of operation.

Trade-off discipline:

Every significant decision in this document includes what was given up and why. What I would do differently, and why. Product management at this level is inseparable from honest accounting of the choices made and the choices avoided.

CognArc · AI Product Management Case Study · Portfolio Document

Agentic Architecture · Human Oversight Built-In · Powered by TRIBE v2